

Student Name: _____

Class Name: **College Algebra FA26 - 014 MW 2:30**Number of Questions: **10**Instructor Name: **Mamani Velasco, Ruben****Question 1 of 10**

Solve.

$$(z - 5)(z + 8) = 0$$

(If there is more than one solution, separate them with commas.)

 $z =$ **Question 2 of 10**Solve for u .

$$6u^2 + 18u = 0$$

Question 3 of 10Solve for v .

$$v^2 + 10v + 25 = 0$$

 $v =$ **Question 4 of 10**Solve for w .

$$5w^2 + 2 = -11w$$

 $w =$

Question 5 of 10

Write the quadratic equation whose roots are 1 and 3, and whose leading coefficient is 2.

Question 6 of 10

Fill in the blank with a number to make the expression a perfect square.

$$w^2 - 2w + \boxed{}$$

$$w^2 - 2w + \boxed{}$$

Question 7 of 10

Solve the quadratic equation by completing the square.

$$x^2 - 2x - 4 = 0$$

(a) Choose the appropriate form and fill in the blanks with the correct numbers.

○ $(x + \underline{\hspace{1cm}})^2 = \underline{\hspace{1cm}}$

○ $(x - \underline{\hspace{1cm}})^2 = \underline{\hspace{1cm}}$

(b) Solve the equation. Simplify your answer as much as possible. If there is more than one solution, separate them with commas.

$$x =$$

Question 8 of 10

Use the quadratic formula to solve for x .

$$3x^2 - 6x - 1 = 0$$

(If there is more than one solution, separate them with commas.)

$x =$

Question 9 of 10

The area of a rectangle is 44 ft^2 , and the length of the rectangle is 3 ft less than double the width.

Find the dimensions of the rectangle.

Length: _____ ft

Width: _____ ft

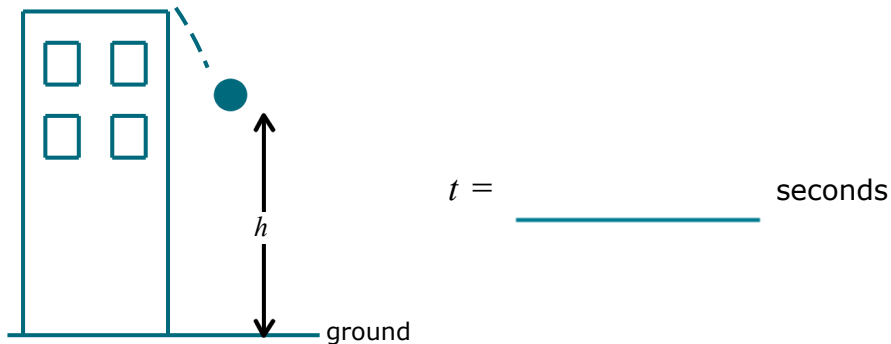
Question 10 of 10

A ball is thrown from a height of 157 feet with an initial downward velocity of 9 ft/s. The ball's height h (in feet) after t seconds is given by the following.

$$h = 157 - 9t - 16t^2$$

How long after the ball is thrown does it hit the ground?

Round your answer(s) to the nearest hundredth.
(If there is more than one answer, use the "or" button.)



Class Name: **College Algebra FA26 - 014 MW**
2:30Number of Questions: **10****Question 1 of 10**

$$z = 5, -8$$

Question 2 of 10

$$u = 0, -3$$

Question 3 of 10

$$v = -5$$

Question 4 of 10

$$w = -\frac{1}{5}, -2$$

Question 5 of 10

$$2x^2 - 8x + 6 = 0$$

Question 6 of 10

$$w^2 - 2w + 1$$

Question 7 of 10

(a) ☐ $(x + \boxed{})^2 = \boxed{}$

☒ $(x - 1)^2 = 5$

(b) $x = 1 + \sqrt{5}, 1 - \sqrt{5}$

Question 8 of 10

$$x = \frac{6 + \sqrt{48}}{6}, \frac{6 - \sqrt{48}}{6}$$

The answer can also be written in the simplified form $x = \frac{3 + 2\sqrt{3}}{3}, \frac{3 - 2\sqrt{3}}{3}$.

Question 9 of 10

Length: 8 ft

Width: 5.5 ft

Question 10 of 10

$t = 2.86$ seconds